

WASP-52b

R-1799

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-52_190917 · created 2026-08-03T01:46:31+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2458743.8213 +/- 0.0051 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.21 +/- 0.062
Transit depth (Rp/Rs)^2	4.42 +/- 2.6 [%]
Orbital Inclination (inc)	84.43 +/- 2.78
Airmass coefficient 1 (a1)	15.91 +/- 10.8
Airmass coefficient 2 (a2)	-2.13 +/- 0.82
Scatter in the residuals of the lightcurve fit is	71.0 %
Best Comparison Star	#1 - [139, 137]
Optimal Aperture	3.8
Optimal Annulus	6.97
Transit Duration (day)	0.067 +/- 0.026

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.21 ± 0.062 vs 0.1646 (deviation 0.44σ)
Duration fit vs published	0.067 d vs 0.0758 d (deviation 0.2σ)
Mid-time O–C	12.4 min
Depth SNR	2.0
Residual scatter	71.0 %

Light curve

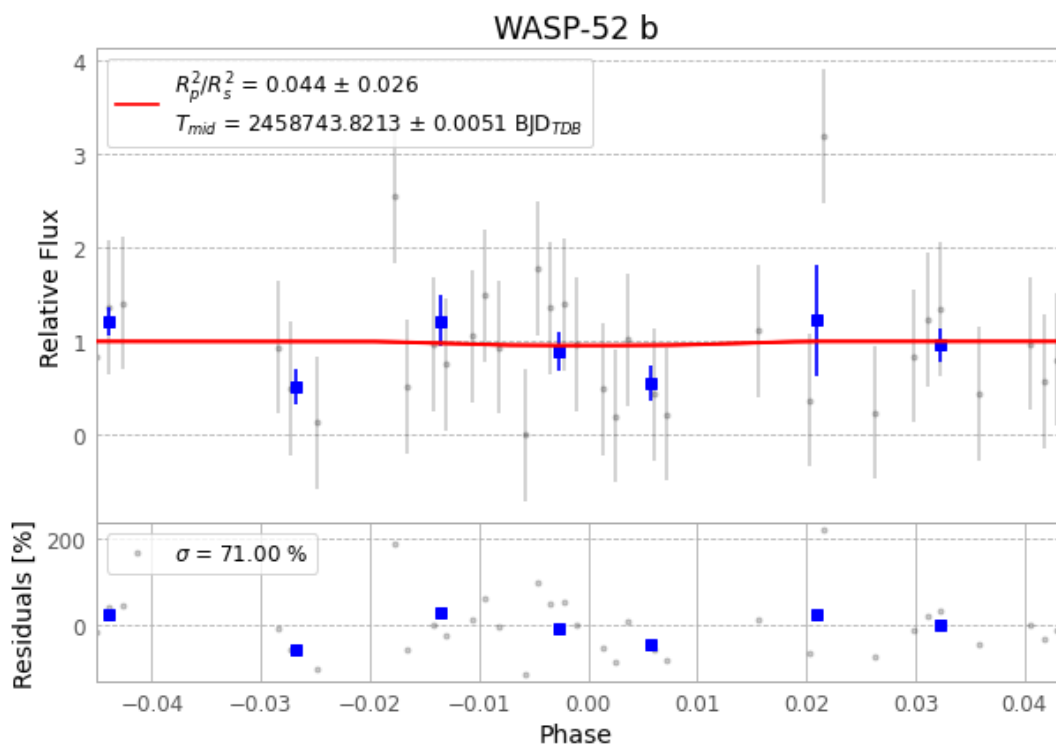


Figure 1 — FinalLightCurve_WASP-52 b_2019-09-16.png (SHA-256 83d93692bc8a6627...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [368, 84], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 b569768293df1c4c...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

77 FITS frames (51 MB), WASP-52190917053312.FITS → WASP-52190917092112.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-52-190917.log (7d1353bc9a4a...), FinalLightCurve_WASP-52 b_2019-09-16.png (83d93692bc8a...), FinalLightCurve_WASP-52 b_2019-09-16.csv (afc7195e58e3...)

Compute resources

Wall time 215.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-03 01:46Z.